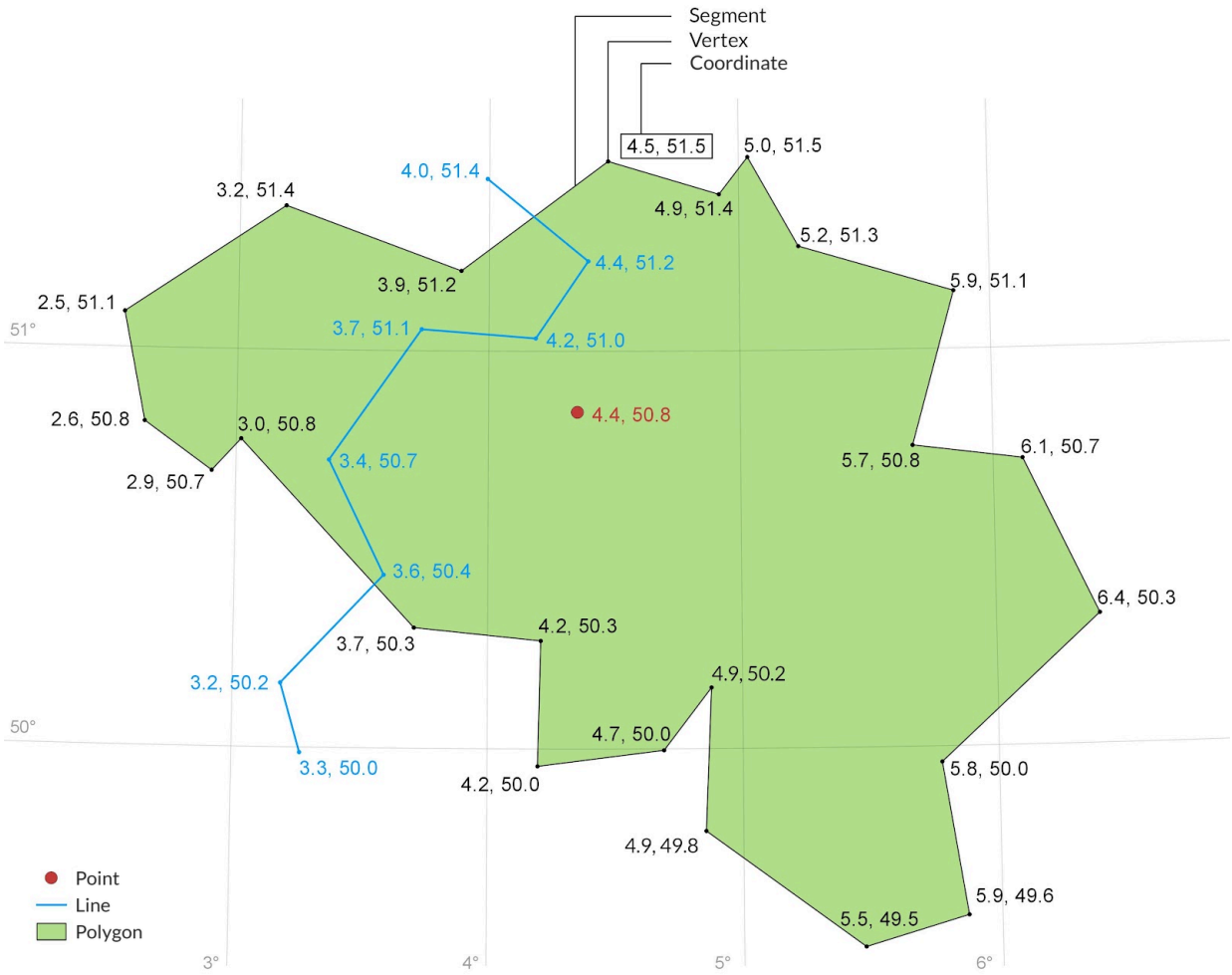
2.2 Vector model

The second commonly used data model in GIS is the **vector model**. It describes geographic information as separate geometric objects (*features*): points, lines and areas – called *polygons*. The shape and position of each object are defined by coordinates: a point by one coordinate pair, a line or a polygon by the coordinates of its (corner) points – called *vertices*. Below, we take a short look at the main characteristics of the vector model and give some examples.

Points, lines and polygons

> See also
>
> The raster model is discussed on Raster model. A comparison of both models is on Rasters vs. vectors.

- **Points** consist of one coordinate.
- **Lines** consist of at least two coordinates (a start point and an end point).
  All points of a line, including the start and end points, are called *vertices*. The parts of the line between two successive vertices are called *segments*.
- **Polygons** consist of a line whose start point and end point are the same.
  The whole area enclosed by this line also belongs to the polygon. A polygon needs at least four coordinates (a triangle): three different points, plus the first point repeated as the end point.



Very accurate

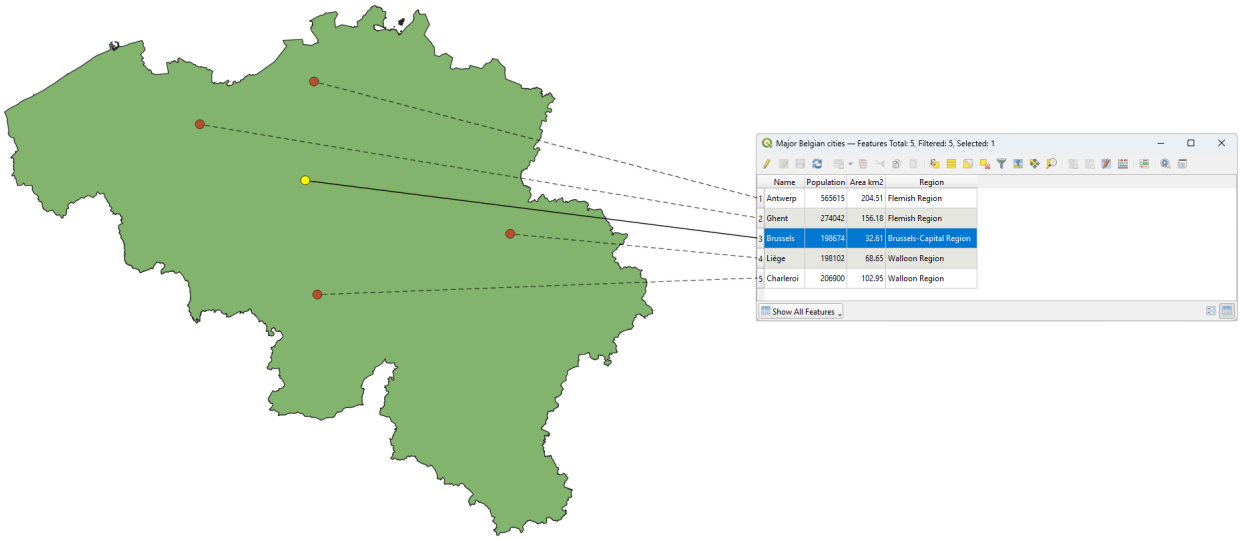
As you learned on the previous page, data in the raster model are positioned in space by means of a grid of cells. Vector objects, on the other hand, are positioned in space by means of **exact coordinates**. So, unlike with rasters, there is no generalisation to a grid of cells. In other words, vectors are **not resolution-dependent**. This also has visual consequences: no matter how far you zoom in on a vector object, the object will always be shown sharply. Technically, there is no limit to the level of detail you can work with. In practice, of course, the accuracy depends on how precisely the data were drawn.

Because vector objects have exact positions and sharp boundaries, they are very suitable for storing **discrete data**, but less suitable for continuous data.

Attribute data

Another important difference from the raster model is that a **variety of attribute data** can be linked to vector objects. Besides numbers (whole or decimal), these can also be text or dates, for example. Moreover, while a raster cell in practice usually holds only one value (or a few), vector objects often have many attribute fields.

The example below shows a vector file with point objects that represent cities. Each **vector object** – also called a **feature** – contains several attribute fields: the city's name (text), the number of inhabitants (a whole number), the area of its territory (decimal number), and the region it belongs to (text).



In QGIS and similar programs, you can consult the attribute data of a single feature directly in the map. You can also view the attribute data of all features of a layer at once, in the form of a table. This **attribute table** has one row per feature and one column per attribute field, and you will use it a lot when working in GIS.

> **i** See also
>
> How to open and use the attribute table in QGIS is explained on The attribute table.

Complex, but small files

Vector files are usually small in size. After all, only the coordinates of the vertices and the attribute data have to be stored. On the other hand, the vector model is a fairly complex data structure. As a result, some spatial analyses may take a lot of computing power and time.